

SAI VENKATESH ATUKURI

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SUMMARY

- Certified Snow Pro with 5+ years of experience in data analysis, design, modeling, development, and testing of Data Warehouse applications.
- Proficient in Snowflake data transformations, SQL optimizations, SnowSQL, Snowpipe, Streams, Time Travel, and Cluster Keys.
- Skilled in managing data pipelines using Snowpipe for seamless data loading and automation with DBT Cloud, including models, macros, and seeds.
- Strong expertise in SQL Server, MySQL, and PostgreSQL for database management and writing complex queries with stored procedures and functions.
- Experienced in handling diverse data formats, including JSON, CSV, and XML, for structured and semi-structured data transformations.
- Expertise in PySpark, Pandas, and NumPy for efficient data processing, manipulation, and analysis of large datasets.
- Expertise in creating interactive dashboards and reports using Tableau, Power BI, and Advanced Excel.
- Well-versed in cloud computing with AWS (EC2, Lambda, S3, RDS, SageMaker) and Azure for scalable and efficient data storage and processing.
- Skilled in ETL processes, data warehousing, and ensuring data quality through efficient workflows.
- Proficient in project management tools like Jira, Trello, and Asana, ensuring seamless collaboration and project execution.
- Experienced in creating and maintaining technical documentation, including TDD, BRD, FRD, and Test Plans, for effective project management.
- Adept in data storytelling, A/B testing, forecasting, clustering, and regression to drive business decisions.

EDUCATION

- **Master of Science in Computer and Information Systems**, Wilmington University, New Castle, DE, USA
- **Bachelor of Science in Electronics & Communication Engineering**, R.V.R & J.C College of Engineering, Guntur, AP, India

TECHNICAL SKILLS

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| • Programming Languages: | Python (Pandas, NumPy, Data Build Tool, Pyspark), Java, SQL, Presto/Trino, PL/SQL |
| • Data Warehousing: | Snowflake, Redshift, Pyspark, Hive |
| • Cloud technologies: | AWS, Azure, DBT Cloud |
| • SQL/NoSQL Platforms: | SQL Server, Google Big Query, MongoDB, MySQL |
| • Development Platforms: | GitHub, Visual Studio, Sublime |
| • Others: | TIBCO Statistica, TIBCO Spotfire, Hadoop, Agile (Scrum), JIRA |

EXPERIENCE

Data Engineer-Vertex Pharmaceuticals, Boston, USA

May 2024 – PRESENT

- Collaborated with clients and design teams to investigate data sources and model requirements, ensuring alignment with business objectives.
- Built pipelines to extract raw CSV and JSON files from AWS S3, transforming them into structured tables in Snowflake's raw schema.
- Developed SnowSQL scripts and implemented Snowpipe for continuous data injection from AWS S3 using event handlers.
- Configured dbt_project.yml for streamlined workflows, defining project settings, schemas, materializations, and environment-specific configurations.
- Created scalable DBT models using advanced SQL, CTEs, and Jinja templating for dynamic, reusable data transformations.
- Designed star schema-based physical data models, including fact and dimension tables, to optimize performance and improve query efficiency.
- Implemented macros for cleansing errors and managing invalid records, enhancing data quality and auditability.
- Built DBT seeds for referential tables, eliminating hard coding, standardizing records, and supporting reporting requirements.
- Automated ICV and UAT validation scripts using macros, reducing manual testing efforts by 40%.
- Extracted reporting views into Power BI and developed interactive dashboards for actionable stakeholder insights.
- Automated deployment pipelines by integrating DBT Cloud with Bitbucket, streamlining TST and PROD releases.
- Modeled Oracle Process Manufacturing (OPM) data into Snowflake, aligning with reporting and business requirements.

Data Engineer, MetLife, NJ, USA

JAN 2023 - April 2024

- Collaborated with cross-functional teams—including stakeholders and product owners to gather data requirements and integrate collected datasets.
- Built Apache Airflow DAGs using custom Python scripts and operators to orchestrate end-to-end ETL workflows from S3 to Snowflake.
- Automated data validation, cleansing, and preprocessing using Python (Pandas, NumPy) to prepare structured and semi-structured datasets.
- Created modular DBT models using CTE and reusable Jinja macros in Snowflake, leveraging Python-driven parameterization for transformation logic.
- Ingested data from Amazon S3 and TIBCO Spotfire via Statistica data workflows into Snowflake for further integration and analytics.
- Automated claims data pipelines by integrating AWS Lambda functions with Snowflake for event-driven transformations.
- Processed semi-structured data (CSV, JSON) using AWS Glue and ingested into Snowflake for downstream analysis.
- Developed DBT models, macros, and Snowflake sequences to automate workflows and handle surrogate keys for data modeling.
- Employed incremental materialization and ephemeral materialization techniques in DBT workflows to optimize data processing and improve efficiency.
- Used Fivetran to load raw source data into Snowflake raw schemas, enabling centralized access for modeling and BI workflows.
- Designed and deployed Power BI dashboards sourced from Snowflake, enabling stakeholders to monitor claims and renewals in real time.
- Authored Functional Requirement Documents (FRDs) detailing data models, reducing rework by 30%.

Data Engineer - TCS, Hyderabad, India

April 2019 - DEC 2021

- Collaborated with stakeholders, product owners, and design teams to gather data requirements and align data architecture with business goals.
- Designed and deployed daily data pipelines using AWS Glue and AWS Lambda to convert raw CSV and JSON files into Parquet format for optimized storage and performance.
- Automated data ingestion and transformation using AWS Lambda triggers and Glue Jobs, storing the processed Parquet files into Amazon S3.
- Implemented pipeline decoupling using Amazon SQS, and configured SNS notifications for job monitoring, alerting, and downstream service integrations.
- Developed (IaC) templates using Python's Troposphere library and AWS CloudFormation, automating deployment of Glue jobs, IAM roles, S3 buckets.
- Developed Power BI dashboards by combining data from AWS Redshift, enabling stakeholders to make informed decisions.
- Implemented real-time streaming pipelines using Kafka and AWS Lambda, enabling real-time data ingestion, and publishing to Redshift and S3.
- Executed complex data transformations and dimensional modeling in AWS Redshift using advanced SQL functions, optimizing for performance.
- Processed high-volume datasets using Hive and PySpark on Amazon EMR, improving batch performance and supporting large-scale analytics use cases.
- Transformed raw data into Parquet format using PySpark on AWS Glue, and loaded it into Amazon S3 for scalable analytics.

CERTIFICATIONS

- SnowPro Core Certification (COF-C02)